

CSE-420

Assignment - 3

1. Write Three address code for the following program:

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$b = \text{func}(c, d+k*y)$

Note: self study 3. Look into theory policy slide

2.

Observe the following code and answer the questions afterwards.

Note: It is not necessary to show any symbol table.

<pre>struct P { char grade; int marks; char section[6]; };</pre>	<pre>struct Q { int courseId; P result; char room[5]; };</pre>	<pre>struct R { char title[8]; Q course; int semester; };</pre>	<pre>struct S { int studentId; R info; char university[12]; };</pre>
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Now, Suppose the following variable is declared:

S studentRecord;

Assume that:

- **int** occupies **4 bytes** and **char** occupies **1 byte**.
- **@studentRecord** denotes the base address of the object **studentRecord**.

- a. Determine** the total offset of the field accessed in the following statement relative to **@studentRecord**:
- y = studentRecord.info.course.result.section[4];**
- b. Write** the equivalent three-address code for the statement above in part a.

Note: Lecture 17-18